HI 5312 Module 5

Tableau Visual Analytics Cheat Sheet

Purpose: A quick reference for building responsible healthcare worksheets and dashboards with the MIMIC-IV course extract.

1. Core Tableau Concepts

Concept	Meaning
Dimension	Categorical field used to segment or label data.
Measure	Numeric field that Tableau aggregates.
Discrete	Creates headers and separate categories.
Continuous	Creates an axis and ordered range.
Marks card	Controls color, size, label, detail, tooltip, and mark type.
Filter	Restricts the displayed data.
Parameter	User-controlled value that drives a calculation or display.
Action	Allows a view to filter, highlight, or navigate another view.

2. Choose the Chart from the Question

Question	Starting chart
Compare categories	Sorted bar chart
Show change over time	Line chart
Show a numeric distribution	Histogram or boxplot
Relate two numeric measures	Scatterplot
Compare two categorical dimensions	Heat map or grouped bar chart
Show a small true part-to-whole	Bar chart or limited pie chart

Avoid 3-D charts, unsorted rankings, excessive color, dual axes without strong justification, and percentages without denominators.

3. Essential Calculated Fields

Weighted hospital mortality rate

```
SUM([Hospital Deaths]) / SUM([Admissions])
```

Mortality display label

```
STR(ROUND(100 * SUM([Hospital Deaths]) / SUM([Admissions]), 1)) + "%"
```

ICU group

```
IF [ICU Admission] = "ICU" THEN "Includes ICU" ELSE "No ICU" END
```

Higher-volume group

```
IF SUM([Admissions]) >= 50 THEN "Higher Volume" ELSE "Lower Volume" END
```

4. Dashboard Blueprint

Area	Recommended content
Top	Specific title, scope, date range, and key filters
First row	KPI cards: admissions, mortality, LOS, ICU percentage
Middle	Primary time trend and main categorical comparison
Bottom	Supporting detail, subgroup comparison, or diagnostic view
Footer	Source, refresh date, definitions, exclusions, and limitations

5. Formatting Checklist

- State the comparison or finding in the title.
- Display units on axes and labels.
- Sort bars when ranking matters.
- Use one consistent meaning for each color.
- Use whitespace to separate sections.
- Keep tooltips concise and informative.
- Test filters across every worksheet.
- Review null values and suppressed categories.
- Test at the intended screen size.
- Include denominator, source, and limitations.

6. Tableau Public and Healthcare Data

Tableau Public publishes workbooks and underlying data to the internet. Do not publish row-level healthcare data. Use the aggregated course extract and follow institutional and course policy. Use Tableau Desktop for local row-level work when permitted.

7. Presentation Framework

What happened? Describe the visible pattern.

Why might it have happened? Offer plausible explanations without claiming causality.

What next? Recommend a specific investigation or operational step.

8. Final Quality Check

Question	Yes/No
Can the primary message be identified within five seconds?	
Are counts and rates clearly distinguished?	
Are denominators and time periods defined?	
Does every dashboard element support the decision?	
Are missing values and exclusions disclosed?	
Could the display mislead through scale, aggregation, or color?	
Does the dashboard protect confidentiality?	